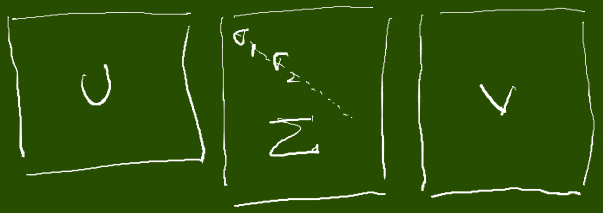
$$\left(A \right) \quad (m, n)$$



$UU^t = I_m \quad \sigma_1 \geq \sigma_2 \geq \ldots$

$VV^t = I_n$

SVD

singular value decomposition

Vectori

$$v = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

K corp comutativ $\quad (K, +, \cdot, 0, 1)$

$(K, +, 0)$ grup comutativ

$(K, \cdot, 1)$ monoid com.

$\forall x \in K \quad x \neq 0$

$\exists x^{-1}$

$x^{-1} \cdot x = x \cdot x^{-1} = 1$

distrib

$x \cdot (y + z) = x \cdot y + x \cdot z$

$$K^n = \left\{ v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_1 \dots x_n \in K \right\}$$

$\mathcal{M}_{n,1}(K)$

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$

$x_n = \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} \longrightarrow \frac{\pi^2}{6}$

$y_n = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{n+1} \cdot \frac{1}{n}$

$\hookrightarrow \frac{\pi}{4}$?

$(\mathbb{Z}_n, +, \cdot)$ corp ? $\Leftrightarrow$ n prim

$\mathbb{Z}_2 = \{0, 1\}$

XOR

+	0	1
0	0	1
1	1	0

Si (AND)

·	0	1
0	0	0
1	0	1